Given Data:

Stainless-Steel 904L	RBSN
L (m) = 0.050	0.010
k (W/m.K) = 13	12
ρ (kg/m^3) = 7,500	2,500
c_p (J/kg.K) = 500	1,100

$\Delta x = 0.005$ m $T_{\infty,2} = 20^\circ C$

$\Delta t = 1$ s $h_2 = 150 (W/m^2.K)$

$T_i(0) = 20^\circ C$

$T_{\infty,1} = 260^\circ C$

$h_1 = 300 (W/m^2.K)$

1. Thermal Diffusion: $\alpha = \frac{k}{\rho . c_p}$

Stainless-steel:

$\alpha_1 = \frac{13}{(7500)(500)} = 3.2911 \times 10^{-6} m^2/s$

RBSN:

$\alpha_2 = \frac{12}{(2,500)(1100)} = 4.3636 \times 10^{-6} m^2/s$

2. Volumetric Heat Capacity

$(\rho . c_p)_1 = (7,900)(500) = 3,950,000 J/m^3.K$

$(\rho . c_p)_2 = (2500)(1100) = 2,750,000 J/m^3.K$

3. Nodal Thermal Capacities: $C_i = \rho . c_p \Delta x$

Node 0: $C_0 = \frac{(3,950,000)(0.005)}{2} = 9875 J/m^2.K$

Node 1-5: $C_{1-5} = (3,950,000)(0.005) = 19,750 J/m^2.K$

Node 6 interface: $C_{6,int} = \frac{(3,950,000)(0.005)}{2} + \frac{(2,750,000)(0.005)}{2} = 16,750 J/m^2.K$

Node 7: $C_7 = (2,750,000)(0.005) = 13,750 J/m^2.K$

Node 8: $C_8 = \frac{(2,750,000)(0.005)}{2} = 6875 J/m^2.K$

4. Thermal Conductances: $G = \frac{k}{\Delta x}$

Steel:

$G_1 = \frac{13}{0.005} = 2,600 W/m^2.K$

RSBN:

$G_2 = \frac{12}{0.005} = 2,400 W/m^2.K$

5. Explicit Finite Difference Equations : $T_i^{n+1} = T_i^n + \frac{\Delta t}{C_i} \sum \dot{q}_{net,i}^n$

For $\Delta t = 1s$:

Node 0:

$$T_0^{n+1} = T_0^n + \frac{1}{9875}\left[300\,(260 - T_0^n) + 2{,}600\,(T_1^n - T_0^n)\right]$$

$$T_0^{n+1} = 0.706329\,T_0^n + 0.263291\,T_1^n + 7.898734$$

Nodes 1-5:

$$T_i^{n+1} = T_i^n + \frac{1}{19750}\left[2{,}600\,(T_{i-1}^n - T_i^n) + 2600\,(T_{i+1}^n - T_i^n)\right]$$

$$T_i^{n+1} = 0.131646\,T_{i-1}^n + 0.736709\,T_i^n + 0.131646\,T_{i+1}^n$$

Node 6:

$$T_6^{n+1} = T_6^n + \frac{1}{16750}\left[2{,}600\,(T_5^n - T_6^n) + 2{,}400\,(T_7^n - T_6^n)\right]$$

$$T_6^{n+1} = 0.155224\,T_5^n + 0.701493\,T_6^n + 0.143284\,T_7^n$$

Node 7:

$$T_7^{n+1} = T_7^n + \frac{1}{13{,}750}\left[2{,}400\,(T_6^n - T_7^n) + 2{,}400\,(T_8^n - T_7^n)\right]$$

$$T_7^{n+1} = 0.174545\,T_6^n + 0.650908\,T_7^n + 0.174545\,T_8^n$$

Node 8:

$$T_8^{n+1} = T_8^n + \frac{1}{6875}\left[2{,}400\,(T_7^n - T_8^n) + 150\,(20 - T_8^n)\right]$$

$$T_8^{n+1} = 0.349091\,T_7^n + 0.629091\,T_8^n + 0.436364$$

6. Stability Calculation: $\Delta t_{max,i} = \dfrac{C_i}{\sum G_i + h_i}$

$\Delta t_{max,0} = \dfrac{9875}{2{,}600 + 300} = 3.405\,s$

$\Delta t_{max,1-5} = \dfrac{19750}{2{,}600 + 2{,}600} = 3.798\,s$

$\Delta t_{max,6} = \dfrac{16750}{2{,}600 + 2400} = 3.350\,s$

$\Delta t_{max,7} = \dfrac{13750}{2400 + 2400} = 2.696\,s$

$\Delta t_{max,8} = \dfrac{6875}{2{,}400 + 150} = 2.696\,s$

$\Delta t = 1s \;\angle\; \Delta t_{max} = 2.696\,s$

7. First Time Step ($t = 0$ to $t = 1s$):

Initial Condition: $T_0^0 = T_1^0 = \cdots = T_8^0 = 20°C$

Node 0:

$$T_0^1 = 20 + \frac{1}{9875}\left[300(260 - 20) + 2600(20-20)\right]$$

$$T_0^1 = 20 + \frac{72{,}000}{9{,}875}$$

$$T_0^1 = 27.2911\ °C$$

Node 1-5:

$$T_i^1 = 20 + \frac{1}{19750}\left[2{,}600(0) + 2600(0)\right]$$

$$T_{1-5}^1 = 20°C$$

Node 6:

$$T_6^1 = 20 + \frac{1}{16750}\left[2{,}600(0) + 2400(0)\right]$$

$$T_6^1 = 20°C$$

Node 7:

$$T_7^1 = 20 + \frac{1}{13750}\left[2{,}400(0) + +2{,}400(0)\right]$$

$$T_7^1 = 20°C$$

Node 8:

$$T_8^1 = 20 + \frac{1}{6{,}875}\left[2{,}400(0) + 150(20-20)\right]$$

$$T_8^1 = 20°C$$

8. Second Time Step ($t = 1$ to $t = 2s$):

Node 0:

$$T_0^2 = 27.2911 + \frac{1}{9875}\left[300(260-27.2911) + 2{,}600(20-27.2911)\right]$$

$$T_0^2 \approx 32.44°C$$

Continue ...

Node 1:

$$T_1^2 = 20 + \frac{1}{19750}\left[21{,}600(27.2911-20) + 2600(20-20)\right]$$

$$T_1^2 \approx 20.960°C$$

Node 2-8: $T_{2-8}^2 = 20°C$

9. Steady-State Target Temperatures – 95%:

$$T_{i,95\%} = T_{i,0} + 0.95\,(T_{i,ss} - T_{i,0})$$

$T_{i,95\%} = 20 + 0.95\,(T_{i,ss} - 20)$ * Calculation made in excel/spreadsheet:

	A	B	C
1	Node	Ti,ss	Ti,95%
2	0	199.122	190.1659
3	1	192.098	183.4931
4	2	185.073	176.8194
5	3	178.049	170.1466
6	4	171.024	163.4728
7	5	164	156.8
8	6	156.976	150.1272
9	7	149.366	142.8977
10	8	141.756	135.6682

10. Numerical Time Results:

Node	Time to 95%(s)
0	981
1	1,033
2	1,076
3	1,110
4	1,136
5	1,156
6	1,169
7	1,177
8	1,180

$t_{95\%,\,explicit} = \max(t_{95\%,i})$

$t_{95\%\,explicit} = 1{,}180\,s$

$$f_8 = \frac{T_8(1180) - 20}{T_{8,ss} - 20}$$

$$f_8 = \frac{135.683 - 20}{141.756 - 20}$$

$f_8 \approx 0.9501 = 95.01\%$

11. Time Step Convergence:

Δt (s):	Time to 95% (s)
2.0	1,178
1.0	1,180
0.5	1,180.0
0.25	1,180.5
0.1	1,180.6
Matrix exponential	1,180.71

12. Final Required Heating time:

$t_{95\%} = 1180.71\,s$

$t_{95\%} = \frac{1180.71}{60} = 19.68\,min$

Limited Node = 8

$T_{8,95\%} = 135.6683\,^{\circ}C$